Lösungen Übungsblatt 6

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① Suppose by contradiction that (x_n) does not converge to L. Then $\exists \varepsilon_0 > 0$ s.t. $\forall N \in \mathbb{N}\ \exists n > N$ with $|x_n - L| \geq \varepsilon_0$. It follows that we can choose a subsequence (x_{n_k}) with $n_1 < n_2 < \dots < n_k < \dots$ s.t. $|x_{n_k} - L| \geq \varepsilon_0\ \forall k \in \mathbb{N}$.

This subsequence is bounded, as (x_n) is bounded $\overset{\text{Bolzano Weierstrass}}{\Longrightarrow}$ we can extract convergent subsequence of (x_{n_k}). This subsequence is a convergent subsequence of (x_n) but it cannot converge to L, contradicting our assumption. It follows that (x_n) must converge to L.

② a) <u>The series diverges:</u>

$\frac{k+1}{k} = 1 + \frac{1}{k} > \frac{1}{k}\ \forall k \in \mathbb{N}, k \geq 1$. Hence $\sum_{k=1}^{\infty} \frac{1}{k} < \sum_{k=1}^{\infty} \frac{k+1}{k}$ and as $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges, so does $\sum_{k=1}^{\infty} \frac{k+1}{k}$ by comparison test.

b) <u>The series converges to $3/5$</u>: $\left|\frac{-2}{3}\right| = \left|\frac{2}{3}\right| < 1$ hence by geometric series $\sum_{k=0}^{\infty} \frac{(-2)^k}{3^k} = \sum_{k=0}^{\infty} q^k = \frac{1}{1-q} = \frac{1}{1-(-2/3)} = \frac{1}{5/3} = \frac{3}{5}$

c) <u>The series converges to $3/2$</u>: $S_N = \sum_{n=2}^{N} \frac{2}{n^2-1} = \sum_{n=2}^{N} \left(\frac{1}{n-1} - \frac{1}{n+1}\right) = 1 + \frac{1}{2} - \frac{1}{N} - \frac{1}{N+1} = \frac{3}{2} - \left(\frac{1}{N} + \frac{1}{N+1}\right)$

Hence, $\sum_{n=2}^{\infty} \frac{2}{n^2-1} = \lim_{N\to\infty} S_N = \lim_{N\to\infty} \frac{3}{2} - \underbrace{\left(\frac{1}{N} + \frac{1}{N+1}\right)}_{\to 0} = \frac{3}{2}$

③ a) The series converges by Ratio test:

with $a_k = \frac{2^k \cdot 3^k}{k!\,(2k+1)}$ we have: $\frac{a_{k+1}}{a_k} = \frac{2^{k+1} \cdot 3^{k+1}}{(k+1)! \cdot (2k+3)} \cdot \frac{k!\,(2k+1)}{2^k \cdot 3^k} = \frac{2 \cdot 3 \cdot (2k+1)}{(k+1)(2k+3)} \xrightarrow{k\to\infty} 0$

b) The series converges by root test:

$\sqrt[k]{\left(\frac{3}{4} + \frac{1}{k}\right)^k} = \frac{3}{4} + \frac{1}{k} \xrightarrow{k\to\infty} \frac{3}{4} < 1$

c) The series converges if $t > e$ (i.e $t \geq 3$) and diverges for $t < e$, i.e for $t = 1$ and $t = 2$:

With $a_k = \frac{k^k}{k! \cdot t^k}$ we have $\frac{a_{k+1}}{a_k} = \frac{(k+1)^{k+1}}{(k+1)! \cdot t^{k+1}} \cdot \frac{k! \cdot t^k}{k^k} = \frac{(k+1)^{k+1}}{k^k \cdot (k+1) \cdot t} = \frac{1}{t}\left(\frac{(k+1)}{k}\right)^k = \frac{1}{t}\left(1 + \frac{1}{k}\right)^k \to \frac{e}{t}$

$\frac{e}{t} < 1$ for $t > e$ and > 1 for $t < e$.

④ a) We have $(a_n - b_n)^2 = a_n^2 + b_n^2 - 2a_nb_n \geq 0$

$\Rightarrow a_nb_n \leq \frac{1}{2}(a_n^2 + b_n^2)$

In particular, $\sum_{n=0}^{\infty} a_nb_n \leq \frac{1}{2}\sum_{n=0}^{\infty}(a_n^2 + b_n^2) = \frac{1}{2}\underbrace{\sum_{n=0}^{\infty} a_n^2}_{<\infty} + \frac{1}{2}\underbrace{\sum_{n=0}^{\infty} b_n^2}_{<\infty} < \infty$

Hence, $\sum_{n=0}^{\infty} a_nb_n$ converges by Comparison test.

b) The statement is true: $S_N = \sum_{n=0}^{N} a_n^2 \leq \left(\sum_{n=0}^{N} a_n\right)^2\ \forall N \geq 1$ (Binomische Formel)

Hence $\sum_{n=0}^{\infty} a_n^2 = \lim_{N\to\infty} \sum_{n=0}^{N} a_n^2 \leq \lim_{N\to\infty}\left(\sum_{n=0}^{N} a_n\right)^2 = \left(\lim_{N\to\infty} \sum_{n=0}^{N} a_n\right)^2 = \left(\underbrace{\sum_{n=0}^{\infty} a_n}_{<\infty}\right)^2 < \infty$

und somit konvergiert $\sum_{n=0}^{\infty} a_n^2$.

But the other direction is not true: take $a_n = \frac{1}{n}$. Then $\sum_{n=0}^{\infty} a_n^2 = \sum_{n=0}^{\infty} \frac{1}{n^2}$ converges, but $\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} \frac{1}{n}$ does not converge.

⑤ a) $\sum_{k=n+1}^{m} (s_k - s_{k-1})c_k = \sum_{k=n+1}^{m} s_kc_k - \sum_{k=n+1}^{m} s_{k-1}c_k$

$= \sum_{k=n+1}^{m} s_kc_k - \sum_{k'=n}^{m-1} s_{k'}c_{k'+1}$

$= s_mc_m - s_nc_n + \sum_{k=n}^{m-1} s_k(c_k - c_{k+1})$

b) Let $s_n = \sum_{i=0}^{n} \alpha_i$, then by assumption there exists $M > 0$ s.t. $|s_n| \leq M\ \forall n \in \mathbb{N}$.

As $c_n \to 0$, we can choose $\varepsilon > 0$ and find $n_0 \in \mathbb{N}$ s.t. $c_n < \frac{\varepsilon}{2M}\ \forall n > n_0$.

Because $s_k - s_{k-1} = \alpha_k$ we have for $m \geq n > n_0$ using a)

$\sum_{k=n+1}^{m} \alpha_k c_k = \sum_{k=n+1}^{m} (s_k - s_{k-1})c_k \overset{a)}{\leq} s_mc_m + s_nc_n + \sum_{k=n}^{m-1} s_k(c_k - c_{k+1})$

$\leq M\left(c_m + c_n + \underbrace{\sum_{k=n}^{m-1}(c_k - c_{k+1})}_{c_n - c_m}\right) = M(c_m + c_n + c_n - c_m) = M \cdot 2c_n < \varepsilon$ as $n > n_0$.

Hence, by Cauchy convergence, $\sum_{k=0}^{\infty} \alpha_k c_k$ converges.

b) If $a_n = \frac{2^n}{(2n)!}$, then $\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{(2(n+1))!} \cdot \frac{(2n)!}{2^n} = \frac{2 \cdot (2n)!}{(2n+2)!} = \frac{2}{(2n+2)(2n+1)} \xrightarrow{n\to\infty} 0$

hence by ratio test $\sum_{k=0}^{\infty} \frac{2^n}{n \cdot (2n)!}$ converges. Now let $c_n = \frac{1}{n}$, then $c_k > c_{k+1} > 0\ \forall k \in \mathbb{N}$ and $\lim_{k\to\infty} c_k = 0$. Hence, by b), we have $\sum_{k=0}^{\infty} \frac{1}{n} \cdot \frac{2^n}{(2n)!}$ converges.